

Experiment resources

Article 9 Risk management system

Zero shot prompting:

Sophie's team was building an AI tool to help hospitals prioritise urgent patient cases, and before anyone was allowed to use it, she gathered the engineers, doctors, nurses, and legal adviser around one table. Instead of asking, "Does the system work?", she asked, "Who could be harmed if it works badly, is misunderstood, or is used in a way we can already predict?" They listed risks to patient safety, unfair outcomes for certain groups, overreliance by busy staff, and mistakes that might appear only after the tool was used in real hospital conditions. For each risk, they decided what could be designed out of the system, what needed a warning or control, and what doctors needed to know before using it. They tested the tool again and again, not only in ideal conditions but also with messy cases, tired users, and unusual patient files. Some risks could not be removed completely, so Sophie wrote down why the remaining risk was acceptable and what safeguards would stay in place. After launch, the team kept watching how the system behaved, collected incident data, reviewed the risk file regularly, and updated the tool whenever new problems appeared. To Sophie, compliance was not a form signed at the end; it was more like keeping a hospital fire plan alive: identify dangers early, reduce them as far as possible, test the precautions, explain the remaining risks clearly, and keep checking for new ones throughout the system's life.

Contextual prompting:

Mila's AI startup had built a tool that helped hospitals decide which patient messages looked urgent, but before launching it, her team treated the system like a new colleague who needed careful supervision, not like a finished product. They first sat around a table and asked what could go wrong if the tool was used exactly as intended: could it miss a serious symptom, make unfair suggestions for certain patient groups, or push nurses to trust it too much? Then they imagined realistic mistakes people might make, like using it for children even though it had only been tested on adults, or relying on it during very busy night shifts without checking the output properly. For every risk they found, they wrote down how serious it could be, how likely it was, and what they could do about it. Some problems were fixed in the design, such as adding warnings when the model was uncertain. Others could not be removed completely, so they added safeguards, like human review for urgent cases, clear instructions for hospital staff, and training notes explaining when the tool should not be used. Before release, they tested the system again and again, including with realistic hospital scenarios, to see whether the safeguards actually worked. Even after the product went live, Mila kept a shared risk log open: if hospitals reported confusing outputs, unexpected behaviour, or new situations the team had not predicted, the team reviewed the risks again and updated the system or its instructions. By the time investors asked whether the company was "AI Act ready," Mila did not point to a single checklist. She pointed to the living process her team had built: identify risks, judge them, reduce them where possible, explain the remaining ones, test before launch, and keep learning after launch. That was their risk management system.

Rephrase and Respond:

Maya worked at a small company building an AI tool that helped hospitals decide which patients should be seen first in the emergency department. At first, the team was excited because the system seemed fast and accurate, but before anyone was allowed to use it, Maya gathered the doctors, nurses, engineers, and patient safety officer around one table and asked a simple question: “What could go wrong, even if everyone uses this exactly as intended?” They listed the obvious dangers first: the AI might underestimate a child’s symptoms, misunderstand rare conditions, perform worse for people whose data was underrepresented, or make staff trust its recommendation too much. Then Maya asked a second question: “How might people misuse it in a way we can reasonably predict?” The team imagined a crowded night shift where a tired nurse might follow the AI score without checking the patient, or a hospital manager might use the tool outside the emergency department even though it was not designed for that. Instead of treating these worries as paperwork, Maya turned them into design decisions. The system was changed so that it could not make the final decision by itself, displayed warnings when cases were uncertain, explained the main reasons behind its recommendation, and clearly told staff when the tool should not be used. The team tested it again and again, including with difficult cases, children’s cases, and examples from different patient groups, until the remaining risks were reduced as much as technically possible and judged acceptable. Every important choice was written down: the risks found, the tests run, the trade-offs made, the safeguards added, and the instructions hospitals would need. When the tool finally went live, Maya did not close the risk file and move on. She set up a routine to review incident reports, user feedback, and performance data from real hospitals, because new risks could appear after deployment. Months later, when the team noticed that the AI was less reliable in a certain type of paediatric case, they updated the system, revised the instructions, and recorded the change.

Article 10 Data governance

Zero shot prompting:

Karen ran a small company that built an AI tool to help hospitals plan patient appointments, but before she let the system learn from old hospital records, she treated the data like ingredients for medicine. She asked where every record came from, why it had first been collected, and whether it was allowed to be used for this purpose. Her team cleaned misspelled entries, checked labels, removed duplicates, and wrote down every assumption they were making, especially what each data point was supposed to represent. When they noticed that rural clinics and older patients were barely present in the training data, Karen stopped the launch, added better examples, and tested whether the system worked fairly for the people who would actually use it. A colleague found that one pattern could accidentally make appointments harder to get for patients with limited mobility, so they changed the dataset and added checks to catch that kind of bias early. Only when the training, validation, and testing data were relevant, representative, as complete and error-free as possible, and suited to the real hospital setting did Karen approve the system, keeping notes so anyone could later understand how the data had been chosen, prepared, checked, and improved.

Contextual prompting:

Mary and Jonas had just launched their Amsterdam AI startup, which built a tool to help hospitals decide which patients should be invited for an early diabetes screening. At first, their model looked impressive: it was fast, accurate in their demo, and investors loved it. But before they offered it to real clinics, Mary stopped the team and said, “We cannot just ask whether the model works. We need to ask whether the data deserves to be trusted.” So they opened a shared document and traced every dataset back to its source: which hospital it came from, why it had originally been collected, whether patients had been informed properly, and whether the data was meant for this kind of prediction. Then they checked how the data had been cleaned, labelled, updated, and combined. One intern noticed that some older records used different names for the same diagnosis, while another found missing values in data from smaller clinics. Instead of ignoring those problems, the team wrote them down and decided how to fix or limit them. Jonas then compared the dataset with the people the system would actually be used for. He realised that most training data came from urban hospitals and did not include enough patients from rural areas, younger adults, or certain migrant communities. “If we train only on the people who are easiest to find,” he said, “the system may fail exactly when it meets people outside that pattern.” The team added more suitable data where they lawfully could, tested whether the model performed differently across groups, and looked for biases that could unfairly delay screening for some patients. When they needed sensitive information to check whether bias existed, they used it only where strictly necessary, protected access carefully, and deleted it once the bias check was done. By the time the startup met its first hospital client, Mary did not simply say, “Our AI is accurate.” She showed a record of the choices they had made: where the data came from, what it was supposed to represent, what gaps they found, how they reduced bias, and why the final training, validation, and testing data were suitable for the medical setting in which the tool would be used. The hospital’s compliance officer smiled and said, “So your data governance is not a folder you created at the end. It shaped the system before patients ever depended on it.”

Rephrase and Respond:

At a city hospital, Mira was asked to help prepare a new AI tool that would support doctors in spotting urgent cases in the emergency department. Before anyone let the system near real patients, she treated the data like ingredients for a meal that many different people would have to eat safely. She did not just gather old patient files and hand them to the developers. First, she asked where every record came from, why it had originally been collected, and whether it was appropriate to use it for this new purpose. Then her team cleaned the files, corrected obvious mistakes, checked the labels, removed duplicates, and wrote down the assumptions they were making, such as what certain symptoms or test results were supposed to represent. When the first model looked accurate, Mira still did not celebrate. She noticed that most of the training data came from large urban hospitals, while the tool would also be used in smaller regional clinics. So she added more suitable examples, checked whether older patients, women, and minority groups were properly represented, and tested whether the system made worse recommendations for any group. When the team found that one group was being flagged less accurately, they paused, investigated the cause, and adjusted the data and process before continuing. By the time the hospital approved the tool, Mira could explain not only that the AI had been trained, tested, and validated, but that its data had been chosen, prepared, checked for bias, and matched to the real setting where the system would be used. That is the basic compliance logic of Article

10: for a high-risk AI system, trustworthy results start with data that is governed carefully, relevant to the intended purpose, sufficiently representative, as complete and error-free as possible, and actively checked for harmful bias.

Article 11 Technical documentation

Zero shot prompting:

Megan's team had almost finished their new AI tool for helping hospitals prioritise equipment maintenance, but before anyone talked about launching it, she opened a folder called "How we know this system is safe enough." Into it, she placed the story of the system: what it was built to do, which data had shaped it, how it had been tested, what risks they had found, which limits the hospital staff needed to know, and what safeguards were built in. Each time the engineers changed something important, Megan made them update the folder too, because a document that only described yesterday's system would be useless tomorrow. When the hospital's compliance officer visited, Megan did not try to explain everything from memory; she handed over the folder, clear enough for someone outside the team to understand how the AI met the legal requirements. Because the tool was part of a regulated medical product, she also made sure the same file covered both the AI Act information and the product-law information, instead of creating two confusing piles of paperwork. Her small start-up used the simplified documentation form meant for SMEs, but she treated it seriously: not as a shortcut around compliance, but as a clearer way to prove it. By launch day, the team understood that technical documentation was not decoration for auditors; it was the system's passport, prepared before the AI entered the real world, kept up to date, and detailed enough for authorities or notified bodies to check whether the high-risk AI system truly complied.

Contextual prompting:

Madison and John, two former UvA data science students, had built an AI tool that helped small clinics decide which patient files needed urgent review. At first, they thought their biggest job was making the model accurate, but their advisor reminded them that a system used in healthcare would not be judged only by what it could do on demo day. "Imagine an inspector walks in six months from now," she said. "Could you show not just that it works, but why it was built this way, what data shaped it, what risks you expected, what tests you ran, and how a clinic is supposed to use it safely?" So before offering the tool to their first clinic, they opened a shared folder they called the system's "passport." In it, they described the purpose of the AI, the main design choices, the data used for training and testing, the performance results, the limits they had found, the risks they had identified, and the safeguards they had added, such as human review before any urgent-care recommendation was acted on. When they changed the model after discovering that it performed less well on older patient records, they did not treat the documentation as finished paperwork; they updated the passport so it still matched the real system. Because they were a small startup, they used the simplified form made for companies like theirs, but they still made sure the explanation was complete enough that an outside authority could follow the trail from idea, to development, to testing, to safe deployment. Later, when a clinic asked whether the tool was "EU AI Act ready," Madison did not send a marketing slide. She sent the technical file. The clinic's compliance officer could see that the team had not merely built an AI product and

hoped for trust afterwards; they had kept a clear, living record that made the system inspectable, accountable, and ready to be assessed before people depended on it.

Rephrase and Respond:

Before Lina's hospital could start using its new AI tool to help doctors spot early signs of illness, she refused to let the team simply plug it in and hope for the best. The software seemed impressive in testing, but Lina knew that trust could not depend on promises alone. She asked the engineers to gather everything that explained how the system worked: what it was designed to do, what data had been used to build and test it, what risks they had found, how they had reduced those risks, how accurate the system was, what limits doctors needed to know, and what should happen if something went wrong. At first, one developer complained that this felt like writing a manual for people who would never read it, but Lina explained that the file was more like the system's passport. Without it, no one outside the team could understand where the AI came from, what it was safe to use for, or whether it followed the required rules. A few weeks later, when the hospital changed the way the tool displayed warnings, Lina made sure the documentation changed too, because an old description of a new system would be useless. When an external assessor later asked how the hospital knew the AI was ready, Lina did not rely on a confident speech or a glossy demo. She opened the documentation and showed the whole trail: the design choices, the tests, the safeguards, the instructions, and the updates. The assessor could follow the story of the system from idea to deployment, and that was exactly the point: technical documentation made the AI understandable, checkable, and accountable before it touched real patients.

Article 14 Human oversight

Zero shot prompting:

At the hospital, Mira's team introduced an AI tool that helped rank urgent scan results, but they never treated it like the doctor in charge. Before using it, the supplier showed them exactly what the tool was built for, where it could make mistakes, and what warning signs to watch for. Mira and two senior nurses were trained to read its recommendations, question them, and notice when the system behaved oddly. One afternoon, the tool marked an elderly patient's scan as low priority, but Mira saw that the patient's symptoms did not fit the usual pattern. Remembering that people can become too trusting of automated suggestions, she paused, reviewed the scan herself, and asked a radiologist to check it too. They found a serious issue and overrode the AI's recommendation. Later, when the tool began giving several unusual rankings, Mira used the clinic's stop procedure and reported the problem instead of letting it continue. In practice, the clinic complied with human oversight not by having a person merely "near" the AI, but by making sure trained people could understand its limits, monitor it, challenge it, override it, and stop it safely whenever health, safety, or rights might be at risk, which reflects Article 14's requirements for high-risk AI systems.

Contextual prompting:

Amber and Noor had just launched their AI startup in Amsterdam, building a tool that helped hospitals prioritise patient appointment requests. At first, the system seemed almost magical:

it read incoming forms, noticed urgent words, compared symptoms, and suggested which patients should be seen first. But before they let a hospital use it, Noor added a dashboard that showed not only the system's recommendation, but also why it had made that suggestion, how confident it was, and whether the case looked unusual. Amber then trained the hospital coordinators to treat the AI like a very fast assistant, not like a final decision-maker. One coordinator, Sara, was given clear authority to question the system, override it, or pause it completely if something felt wrong. On the first week, the AI marked an elderly patient's request as "low priority" because the form was short and calm. Sara noticed the warning panel showing that the system had little information and that short descriptions could reduce accuracy. Instead of accepting the recommendation, she opened the patient's file, saw a history of heart problems, and moved the appointment up. Later that day, she reported the issue to Amber and Noor, who adjusted the instructions and added a stronger alert for cases with too little information. The AI had helped sort hundreds of requests, but it had not been allowed to run the process on its own. A person who understood its strengths and weaknesses stayed in charge, watched for mistakes, resisted the temptation to trust it automatically, and could step in before the system's recommendation harmed someone.

Rephrase and Respond:

At a busy hospital, Sofia worked as a senior nurse in a department that had recently started using an AI system to help assess which patients might need urgent attention first. The system was fast and often useful, but the hospital never treated it like a replacement for medical judgment. Before Sofia and her colleagues were allowed to use it, they were trained on what the system could do, where it could make mistakes, and which signs might show that something was wrong with its recommendations. Each morning, Sofia checked the AI dashboard while also speaking with patients, reading their files, and listening to the doctors. One afternoon, the system marked an elderly man as low priority, but Sofia noticed that his breathing looked unusual and that his symptoms did not fit the system's ordinary pattern. Instead of simply accepting the AI's output, she paused, reviewed the information, and called a doctor to examine him immediately. She also recorded why she had disagreed with the system's recommendation. Later that day, when the AI began producing several strange results in a row, Sofia used the hospital's procedure to suspend its use and alerted the technical team. The hospital did not see this as a failure of the AI project; it saw it as the system working as it should, with people remaining informed, attentive, and able to intervene. In practice, human oversight meant that the AI could support decisions, but it could not quietly take control of them.